LEARNER GUIDE

Slide it across

Quadratics. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

Last lesson you stretched and flipped the parabola. Today you slide it sideways. The rule $y = (x - p)^2$ looks almost like $y = x^2$, but the x is replaced by $x - p$. A table shows what that does: every height is exactly the same as $y = x^2$ — the whole curve just moves left or right. Here is the surprise worth slowing down for: $y = (x - 2)^2$ slides to the right, not the left. Why? The lowest point of the curve happens when the bracket is 0, and $x - 2 = 0$ when $x = 2$ — so the bottom, which sat at $x = 0$, now sits at $x = 2$. So the turning point moves to (p, 0) and the line of symmetry becomes $x = p$. You will slide the curve with a table, move a live dial and watch the turning point and line of symmetry travel with it, find the turning point from an equation, and — as a stretch — write the equation that puts the lowest point where you choose. There is no calculator today. The question you are exploring is: where does the curve's lowest point go, and why?

What you are learning to notice

- That $y = (x - p)^2$ is $y = x^2$ slid sideways — same shape, new position.
- That a positive p slides the curve to the right, and a negative p to the left.
- That the turning point is at (p, 0) and the line of symmetry is $x = p$.
- That the reason is the bracket: it is 0 at the lowest point, and $x - p = 0$ when $x = p$.

Moving through it, one step at a time

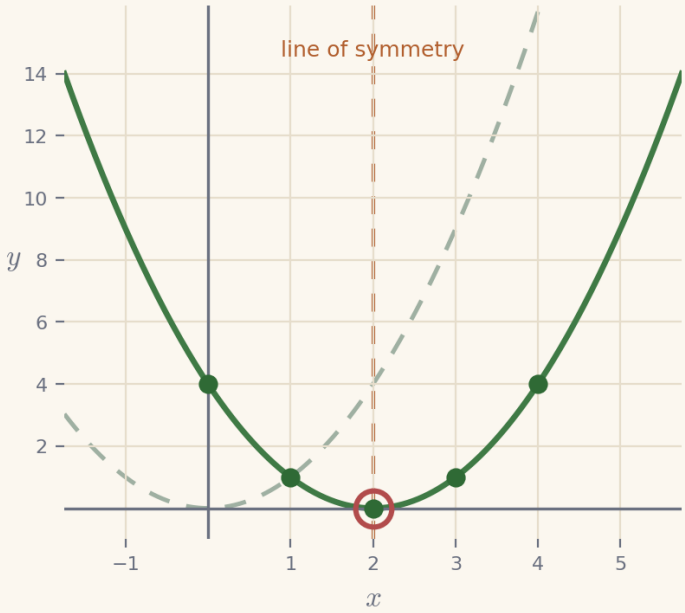
In The same curve, moved sideways, fill a table for $y = (x - 2)^2$ and plot it beside $y = x^2$. In The p-slider, move the dial and watch the curve, its turning point and its line of symmetry slide together. In Find the bottom, read the turning point from an equation. In Write the equation (a stretch), choose p so the lowest point lands where you want.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — slide it right — $y = (x - 2)^2$

Make a table for $y = (x - 2)^2$ by working out the bracket first, then squaring. The dashed curve is $y = x^2$ for comparison.

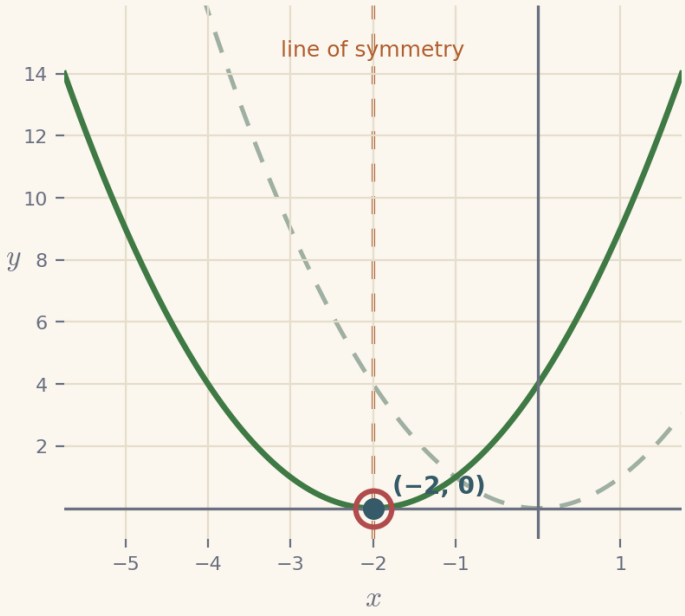


1.	$x = 0$	$(0 - 2)^2 = (-2)^2 = 4$
2.	$x = 2$	$(2 - 2)^2 = 0^2 = 0$
3.	$x = 4$	$(4 - 2)^2 = 2^2 = 4$
4.	lowest point	where the bracket is 0: $x = 2$

the curve is $y = x^2$ slid 2 to the right; turning point (2, 0), line of symmetry $x = 2$

Example 2 — write the equation for a lowest point

Find the rule $y = (x - p)^2$ whose lowest point is at (–2, 0). The bracket must be 0 there.



1. need the bracket 0 at $x = -2$
2. so the bracket is $x - (-2) = x + 2$
3. the rule is $y = (x + 2)^2$
4. check at $x = -2$ $(-2 + 2)^2 = 0$

$y = (x + 2)^2$, with turning point (–2, 0) and line of symmetry $x = -2$

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (plotting $y = x^2$ (Lesson 5), substituting into a bracket, and squaring including squaring a negative), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

NEO by Nudge Education · nudgeeducation.online · Curriculum by Gerry Docherty. Learner Guide v0.1.